

#MOSFET

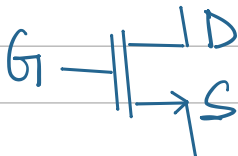
↳ Metal Oxide Semiconductor Field Effect Transistor.

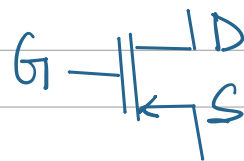
2 types

↳ nmos → Negative channel MOSFET

↳ pmos → Positive channel MOSFET

→ 3 terminal voltage controlled device

Enhancement mode NMOS → 

Enhancement mode PMOS → 

G - Gate, D - Drain, S - Source
controlled terminal → G

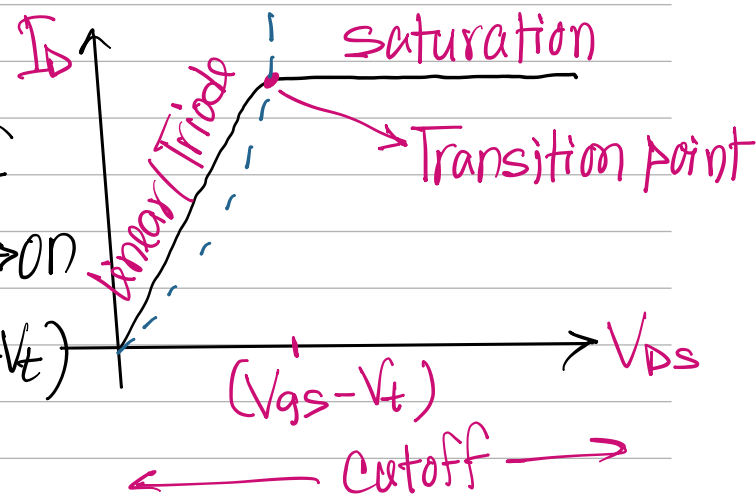
NMOS

V_t → threshold voltage

$V_{gs} < V_t$ → no channel → off

$V_{gs} \geq V_t$ → channel created → on

at transition point $V_{ds} = (V_{gs} - V_t)$



Operating Modes

i) Cutoff → $(V_{gs} < V_t) \rightarrow I_D = 0$

ii) Linear / Triode → $V_{ds} < (V_{gs} - V_t)$

$$\rightarrow I_D = k_n \{ 2(V_{gs} - V_t)V_{ds} - V_{ds}^2 \}$$

iii) Saturation → $V_{ds} > (V_{gs} - V_t)$

$$\rightarrow I_D = k_n (V_{gs} - V_t)^2$$

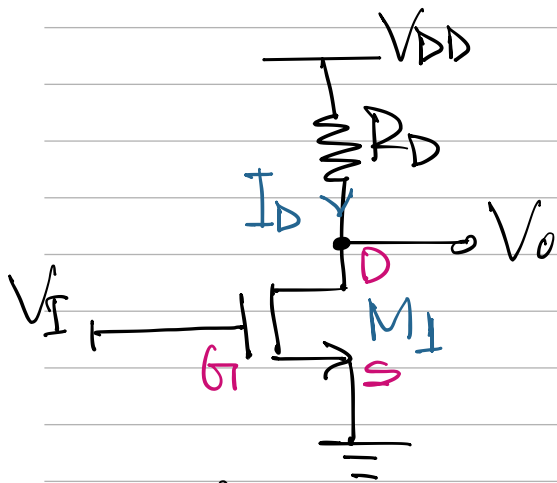
k_n → Process Param
($\mu A/V^2$ or mA/V^2)
will be given

In logic circuits -

Nmos off $\rightarrow$ Nmos will operate in cutoff mode

Nmos on $\rightarrow$ Nmos will operate in linear mode

Nmos Resistive Load Inverter:



Given -

$$K_n = 0.3 \text{ mA/V}^2$$

$$V_{tn} = 0.5 \text{ V}$$

$$R_D = 20 \text{ k}\Omega$$

- i) if $V_{DD} = 2.5 \text{ V} \rightarrow$ find V_O for all input cases.
- ii) P_T , P_{avg}
- iii) find the transition point.

Soln: (i)

here $V_I = V_g = V_{gs}$ and $V_O = V_D = V_{ds}$

Case-1: $V_I = 0 \text{ V} \rightarrow V_{gs} = 0 \text{ V} < V_{tn} = 0.5 \text{ V}$

$\therefore M_1 \rightarrow$ cutoff

$$\therefore I_D = 0$$

$$\Rightarrow \frac{2.5 - V_O}{R_D} = 0 \Rightarrow \boxed{V_O = 2.5 \text{ V}}$$

$$\text{power} \rightarrow P_1 = (2.5 - 0) \times I_D = 0 \text{ mW}$$

Case-II

$$V_I = 5V \quad \therefore M_1 \rightarrow \text{on (linear)}$$

$$I_D = K_n \{ 2(V_{GS} - V_{tn})V_{DS} - V_{DS}^2 \}$$

$$\text{again } I_D = \frac{V_{DD} - V_O}{R_D} = \frac{2.5 - V_O}{20K}$$

equating these two equations —

$$\frac{2.5 - V_O}{20K} = K_n \{ 2(V_{GS} - V_{tn})V_{DS} - V_{DS}^2 \}$$

$$\Rightarrow \frac{2.5 - V_O}{20K} = 0.3 \text{ mA/V}^2 \{ 2(V_I - V_{tn})V_O - V_O^2 \}$$

$$\Rightarrow 2.5 - V_O = 0.3 \times 20 \{ 2(2.5 - 0.5)V_O - V_O^2 \}$$

$$\Rightarrow 2.5 - V_O = 6(4V_O - V_O^2)$$

$$\Rightarrow 2.5 - V_O = 24V_O - 6V_O^2$$

$$\Rightarrow 6V_O^2 - 25V_O + 2.5 = 0$$

$$\text{Solving yields } \rightarrow V_O = 4.064, 0.102V$$

$$\therefore \text{Drain current, } I_D = \frac{V_{DD} - V_O}{R_D} = \frac{2.5 - 0.102}{20K}$$

$$I_D = 0.119 \text{ mA}$$

$$\text{max power consumption } \rightarrow P_2 = P_{\max} = (V_{DD} - 0)I_D$$

$$= 2.5 \times 0.119$$

$$= 0.297 \text{ mW}$$

(ii)

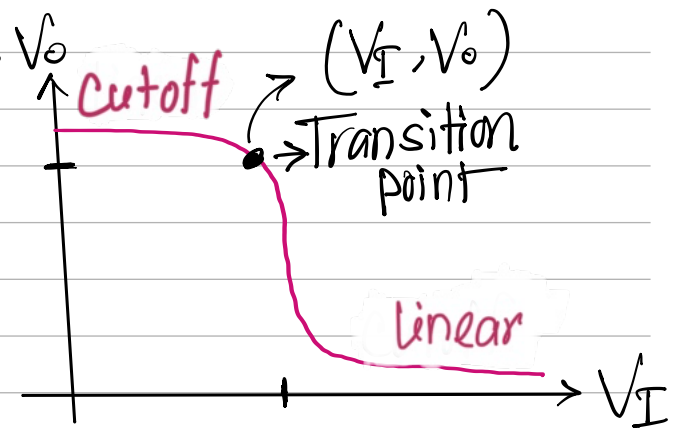
$$\therefore P_T = P_1 + P_2 = 0 + 0.297$$

$$= 0.297 \text{ mW}$$

$$P_{\text{Avg}} = \frac{P_T}{2} = 0.1485 \text{ mW}$$

(iii) Transition Point

here $V_{DS} = V_{GS} - V_{tn}$
 $\Rightarrow V_O = V_I - V_{tn}$



$$\frac{V_{DD} - V_O}{R_D} = K_n \{ 2(V_{GS} - V_{tn})V_{DS} - V_{DS}^2 \}$$

$$\Rightarrow \frac{V_{DD} - V_O}{R_D} = K_n \{ 2V_{DS} \cdot V_{DS} - V_{DS}^2 \}$$

$$\Rightarrow \frac{V_{DD} - V_O}{R_D} = K_n V_{DS}^2$$

$$\Rightarrow \frac{V_{DD} - V_O}{R_D} = K_n V_O^2$$

$$\Rightarrow \frac{2.5 - V_O}{20K} = 0.3 V_O^2$$

$$\Rightarrow 6V_O^2 + V_O - 2.5 = 0$$

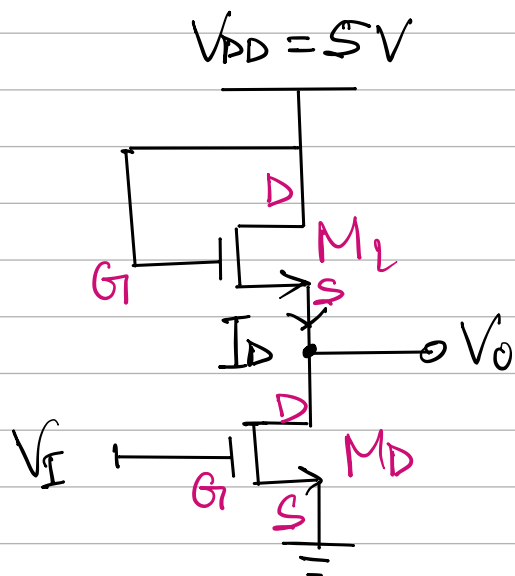
$$V_O = 0.567V, -0.731V$$

$\therefore$ at transition point $\rightarrow V_O = 0.567V$

and $V_O = V_I - V_{tn} \Rightarrow V_I = V_O + V_{tn}$
 $= 0.567 + 0.5$

$$\therefore V_I = 1.067V$$

Nmos Enhancement Load Inverter



Load (M_L)

$$V_{TNL} = 1V, K_{NL} = 10 \mu A/V^2$$

Driver (M_D)

$$V_{TND} = 1V, K_{ND} = 50 \mu A/V^2$$

Determine the V_O , I_D and power dissipation for both input cases.

Soln:

Load MOSFET (M_L)

gate and Drain is connected

$$\therefore V_g = V_D \rightarrow V_{gs} = V_{ds}$$

$$\therefore (V_{gs} - V_t) < V_{ds} \rightarrow \text{always}$$

$\therefore$ Load M_L will be always in Saturation

But Driver $M_D \rightarrow$ either cutoff or linear

Case-I

$$V_I = 0V \therefore M_D \rightarrow \text{cutoff}, M_L \rightarrow \text{saturation}$$

$$\therefore I_D = 0$$

$$I_{DL} = I_D$$

$$\Rightarrow K_{NL} (V_{gsL} - V_{tL})^2 = 0$$

$$\Rightarrow K_{NL} (5 - V_O - 1)^2 = 0$$

$$\Rightarrow (4 - V_O)^2 = 0 \Rightarrow V_O = 4V$$

$$I_D = 0 \therefore P_1 = (V_{DD} - 0)I_D = 0 \text{ mW}$$

Load

$$\begin{aligned} V_{gs} &= V_g - V_s \\ V_{gs} &= 5 - V_O \end{aligned}$$

Case-II

$V_I = 5V$, $M_D \rightarrow \text{Linear}$, $M_L \rightarrow \text{Saturation}$

Driver $\rightarrow I_D = K_{nD} \{ 2(V_{GS_D} - V_{tD}) V_{DS_D} - V_{DS_D}^2 \}$
 $\rightarrow V_{GS_D} = V_I$, $V_{DS_D} = V_o$, $V_{tD} = 1V$

$$\therefore I_D = K_{nD} \{ 2(5-1)V_o - V_o^2 \}$$
$$I_D = 50 \mu A/V^2 (8V_o - V_o^2)$$

Load (M_L) $\rightarrow I_L = K_{nL} (V_{GS_L} - V_{tL})^2$

$$\rightarrow V_{GS_L} = V_{G_L} - V_{S_L} = 5 - V_o, V_{tL} = 1V$$

$$I_L = 10 \mu A/V^2 (5 - V_o - 1)^2$$

$$\Rightarrow I_L = 10 \mu A/V^2 (4 - V_o)^2$$

here, $I_L = I_D$

$$\Rightarrow 10 (4 - V_o)^2 = 50 (8V_o - V_o^2)$$

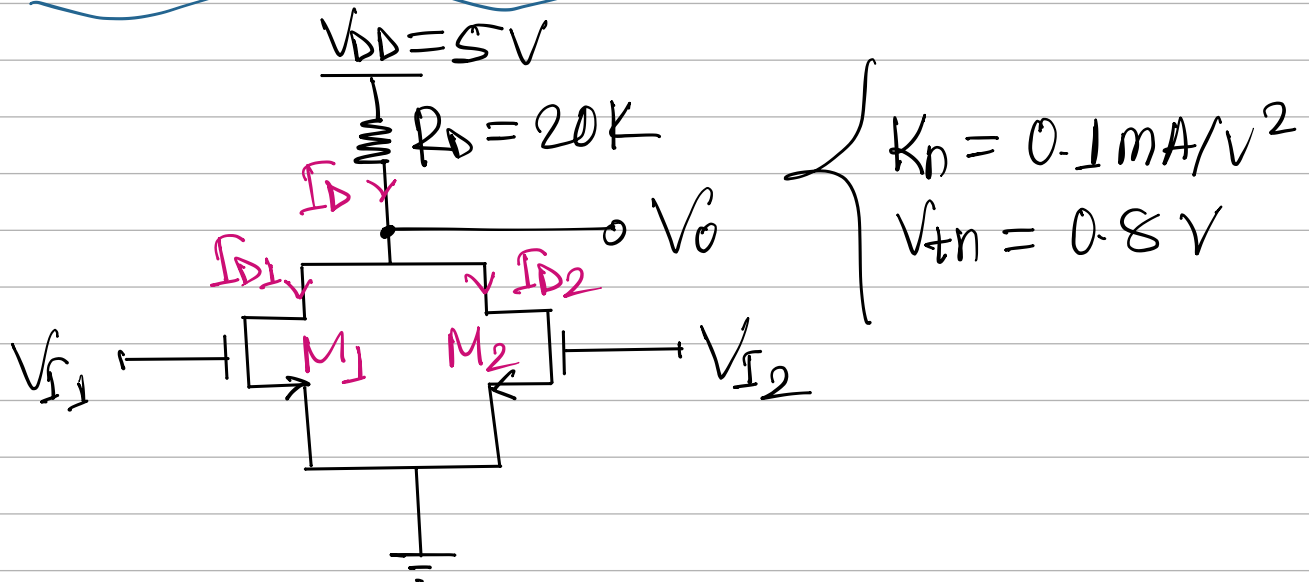
$$\Rightarrow (4 - V_o)^2 = 40V_o - 5V_o^2$$

Solving yields $\rightarrow V_o = 0.348V, 7.56V$

$$\therefore I_L = I_D = 10 \mu A/V^2 (4 - V_o)^2$$
$$= 10 \times 10^{-6} (4 - 0.348)^2 A$$
$$= 0.134 mA$$

$$\therefore P_2 = (V_{DD} - 0) I_D$$
$$= (5 - 0) \times 0.134 = 0.67 mW$$

NMOS Resistive Load NOR gate



- i) find V_O, I_D, I_{D1}, I_{D2} for all cases
- ii) find P_T, P_{avg}

Soln:

Case-I

$$V_{I1} = V_{I2} = 0V$$

$\therefore M_1, M_2 \rightarrow \text{Cutoff}$

$$\therefore I_{D1} = 0, I_{D2} = 0$$

$$\therefore I_D = I_{D1} + I_{D2} = 0$$

$$\Rightarrow \frac{5 - V_O}{R_D} = 0 \Rightarrow V_O = 5V$$

$$\therefore P_1 = V_{DD} \times I_D = 0$$

Case-II

$$V_{I1} = 0V, V_{I2} = 5V$$

$M_1 \rightarrow \text{Cutoff}, M_2 \rightarrow \text{Linear}$

$$I_{D1} = 0$$

$$\text{here, } I_D = I_{D2}$$

$$I_D = \frac{V_{DD} - V_O}{R_D}$$

$$I_{D2} = K_n \left[\frac{1}{2} (V_{GS} - V_{thn}) V_{DS} - \frac{V_{DS}^2}{2} \right]$$

$$\therefore \frac{5-V_0}{20} = 0.1 \{ 2(5-0.8)V_0 - V_0^2 \}$$

$$\Rightarrow 5-V_0 = 2(8.4V_0 - V_0^2)$$

$$\Rightarrow 5-V_0 = 16.8V_0 - 2V_0^2$$

$$\Rightarrow 2V_0^2 - 17.8V_0 + 5 = 0$$

$$\Rightarrow V_0 = 0.290V, \quad 8.60V$$

$$\therefore I_D = I_{D2} = \frac{5-0.29}{20K} = 0.235mA$$

$$\therefore P_2 = (V_{DD}-0) \times I_D = 5 \times 0.235 = 1.1775mW$$

Case-III

$$V_{S1} = 5V, \quad V_{S2} = 0V$$

$M_1 \rightarrow \text{linear}, M_2 \rightarrow \text{cutoff}$

Same as (II) just $\rightarrow I_{D2} = 0, I_D = I_{D1}$

Case-IV

$$V_{S1} = V_{S2} = 5V$$

$\therefore M_1, M_2 = \text{linear}$

$$\text{here } I_D = I_{D1} + I_{D2}$$

$$\Rightarrow \frac{V_{DD}-V_0}{R_D} = 2 \times [K_n \{ 2(V_{GS}-V_t)V_{DS} - V_{DS}^2 \}]$$

$$\Rightarrow \frac{5-V_0}{20K} = 2 \times [0.1 \{ 2(5-0.8)V_0 - V_0^2 \}]$$

$$\Rightarrow 5-V_0 = 4(8.4V_0 - V_0^2)$$

$$\Rightarrow 4V_0^2 - 34.6V_0 + 5 = 0$$

$$V_0 = 0.147V, \quad 8.50V$$

$$\therefore I_D = \frac{5-0.147}{20} = 0.242mA, \quad I_{D1} = I_{D2} = \frac{I_D}{2} = 0.121mA$$

$$\text{power, } P_4 = (5-0)I_D = 1.21mW, \quad P_T = \sum P, P_{Avg} = \frac{P_T}{4}$$

NMOS Depletion type load

The depletion type loads are just another NMOS connected in replacement of R_D . Just the fact that they are of depletion type means their $V_{TN} < 0$. Since they do not need any gate voltage to turn on, V_{GS} can be made 0, by shorting their gate with their source.

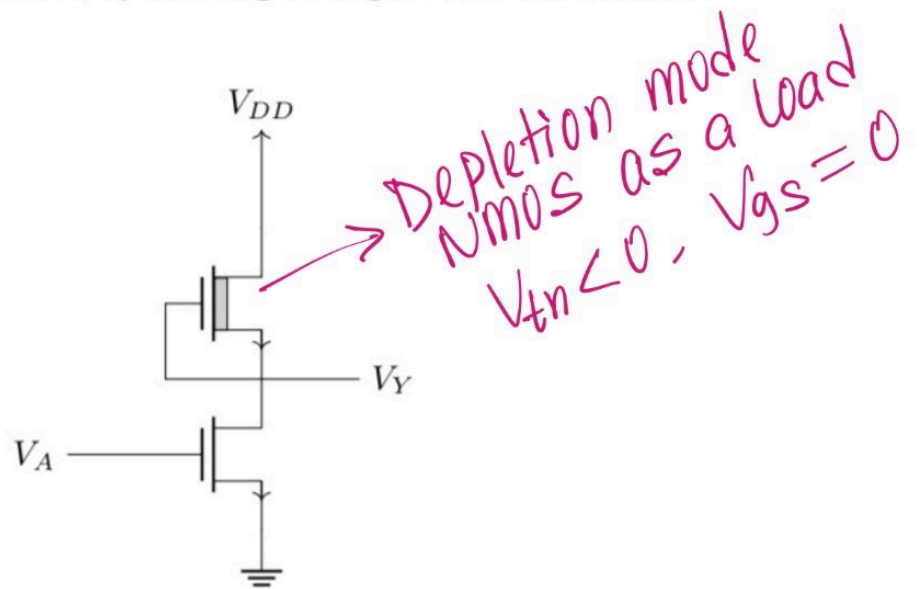
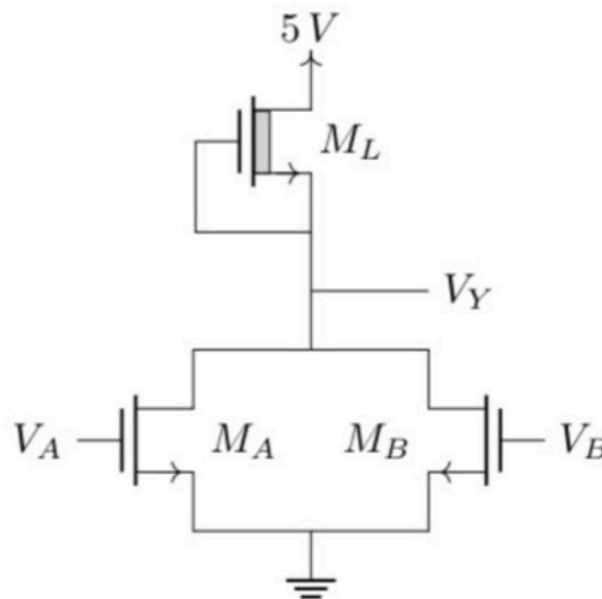


Figure 9.12: A typical NMOS Inverter with depletion type load

NMOS Depletion load NOR gate



process parameter $k_n = k_n' (w/L)$

Problems

It is given that $K'_n = 100 \frac{\mu A}{V^2}$, $V_{TN_A} = V_{TN_B} = 0.4 V$, $V_{TN_L} = -0.6 V$, $\left(\frac{W}{L}\right)_L = 4$, $\left(\frac{W}{L}\right)_A = \left(\frac{W}{L}\right)_B = 1$. **Note:** $K_n = K'_n \left(\frac{W}{L}\right)$. Find all the currents and the output voltage for all cases.

Solution:

Case (0,0): M_A & M_B are both off. Thus, $I_{DS_L} = I_{DS_A} = I_{DS_B} = 0$. Assuming triode mode for M_L ,

$$\begin{aligned} I_{DS_L} = 0 &= K_{n_L} [2(V_{GS_L} - V_{TN_L})V_{DS_L} - V_{DS_L}^2] \\ \Rightarrow K'_n \left(\frac{W}{L}\right)_L [2(0 - (-0.6))(5 - V_Y) - (5 - V_Y)^2] &= 0 \\ \Rightarrow 100 \times 10^{-6} \times 4 [2(0 - (-0.6))(5 - V_Y) - (5 - V_Y)^2] &= 0 \\ \Rightarrow V_Y = 5 V, 3.8 V \end{aligned}$$

Since no current flows, there will be no drops across M_L . So, $V_Y = 5 V$ is the more acceptable answer.

Case (0,1) and (1,0): Both cases will give the same result as M_A & M_B have identical parameters. Let's analyze case (0,1). M_A will thus be off. Since, $V_{OV_L} = V_{GS_L} - V_{TN_L} = 0 - (-0.6) = 0.6 < 5$, M_L might be in saturation mode. Also, as V_Y is expected to be low in this case, we can assume saturation mode for M_B . So, assuming saturation mode for M_L and triode mode for M_B ,

$$\begin{aligned} 100 \times 10^{-6} \times 4(0 - (-0.6))^2 &= 100 \times 10^{-6} \times 1[2(5 - 0.4)V_Y - V_Y^2] \\ V_Y &= 9.04 V, 0.156 V \end{aligned}$$

9.04 V is unacceptable, as it indicates a negative drain current. So, $V_Y = 0.156$. This aligns with our assumption too.

$$\therefore I_{DS_L} = I_{DS_B} = 100 \times 10^{-6} \times 4(0 - (-0.6))^2 = 0.144 mA$$

Case (1,1): In this case, both M_A & M_B can be assumed to be in triode mode, and M_L can be assumed to be in saturation. So,

$$\begin{aligned} 100 \times 10^{-6} \times 4(0 - (-0.6))^2 &= 2 \times 100 \times 10^{-6} \times 1[2(5 - 0.4)V_Y - V_Y^2] \\ \Rightarrow V_Y &= 9.12 V, 0.078 V \end{aligned}$$

9.12 V is unacceptable due to the same reason. So, $V_Y = 0.078 V$. This also aligns with our assumption.

$$\therefore I_{DS_L} = 100 \times 10^{-6} \times 4(0 - (-0.6))^2 = 0.144 mA$$

$$\therefore I_{DS_A} = I_{DS_B} = \frac{I_{DS_L}}{2} = 0.072 mA$$